

DBMS – SELLER AND INVENTORY MANAGEMENT

Project Title: Myntra – E-Commerce database Management System Skill

1. Database Design

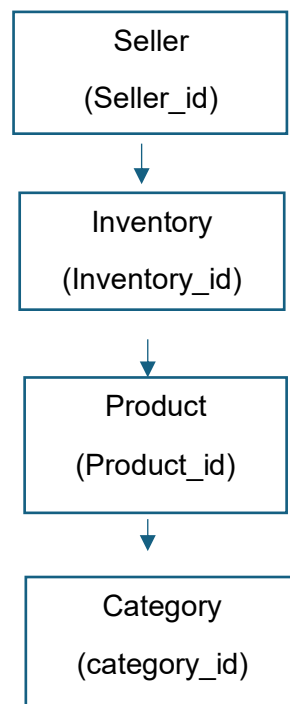
1.1 Seller Table

Attribute	Data Type	Key	Description
Seller_ID	INT	PK	Unique seller ID
Seller_Name	VARCHAR(100)	-	Seller name
Contact_Person	VARCHAR(50)	-	Contact person
Phone	VARCHAR(15)	-	Phone
Email	VARCHAR(100)	-	Email
Address	TEXT	-	Address
GST_Number	VARCHAR(20)	-	GST number

1.2 Inventory Table

Attribute	Data Type	Key	Description
Inventory_ID	INT	PK	Unique inventory ID
Product_ID INT FK References Product	INT	FK	References Product
Quantity	INT	-	Available quantity
Warehouse	VARCHAR(100)	-	Warehouse
Last_Updated	DATE	-	Last updated

2. Table Relationship



3. Table Creation

3.1 Create Seller Table

```
SQL> CREATE TABLE Seller (  
2     Seller_ID INT PRIMARY KEY,  
3     Seller_Name VARCHAR2(100) NOT NULL,  
4     Contact_Person VARCHAR2(50),  
5     Phone VARCHAR2(15),  
6     Email VARCHAR2(100),  
7     Address VARCHAR2(200),  
8     GST_Number VARCHAR2(20)  
9 );
```

Table created.

3.2 Create Inventory Table

```
SQL> CREATE TABLE Inventory (  
2     Inventory_ID INT PRIMARY KEY,  
3     Product_ID INT NOT NULL,  
4     Seller_ID INT NOT NULL,  
5     Quantity INT NOT NULL,  
6     Warehouse VARCHAR2(100),  
7     Last_Updated DATE NOT NULL,  
8     FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID),  
9     FOREIGN KEY (Seller_ID) REFERENCES Seller(Seller_ID)  
10 );
```

Table created.

4. Seller Data Insertion

```
SQL> INSERT INTO Seller  
2 VALUES (301, 'Levis Suppliers', 'Arun Kumar',  
3         '9876543210', 'levissuppliers@gmail.com',  
4         'Chennai', 'GSTINCHN301');  
1 row created.  
  
SQL> INSERT INTO Seller  
2 VALUES (302, 'Vero Moda Suppliers', 'Priya Sharma',  
3         '9876543211', 'veromoda@gmail.com',  
4         'Chennai', 'GSTINCHN302');  
1 row created.  
  
SQL> INSERT INTO Seller  
2 VALUES (303, 'Nike Sports Suppliers', 'Rahul Kumar',  
3         '9876543212', 'nikesports@gmail.com',  
4         'Coimbatore', 'GSTINTN303');  
1 row created.  
  
SQL> INSERT INTO Seller  
2 VALUES (304, 'Fossil Watch Suppliers', 'Karthik Raj',  
3         '9876543213', 'fossilwatches@gmail.com',  
4         'Bangalore', 'GSTINKA304');  
1 row created.  
  
SQL> INSERT INTO Seller  
2 VALUES (305, 'Wildcraft Suppliers', 'Divya Ravi',  
3         '9876543214', 'wildcraftsuppliers@gmail.com',  
4         'Chennai', 'GSTINCHN305');  
1 row created.  
  
SQL> INSERT INTO Seller  
2 VALUES (306, 'Fashion Distributors', 'Meena Devi',  
3         '9876543215', 'fashiondistributors@gmail.com',  
4         'Madurai', 'GSTINTN306');
```

```

SQL> SELECT * FROM Seller;

SELLER_ID
SELLER_NAME
CONTACT_PERSON          PHONE
EMAIL
ADDRESS
GST_NUMBER
301
SELLER_ID
SELLER_NAME
CONTACT_PERSON          PHONE
EMAIL
ADDRESS
GST_NUMBER
Levis Suppliers
SELLER_ID
SELLER_NAME
CONTACT_PERSON          PHONE
EMAIL
ADDRESS
GST_NUMBER
Arun Kumar              9876543210
SELLER_ID
SELLER_NAME
CONTACT_PERSON          PHONE
EMAIL
ADDRESS

```

5. Inventory Data Insertion

```

SQL> INSERT INTO Inventory
  2 VALUES (401, 201, 301, 50, 'Chennai Warehouse', '01-SEP-2026');

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (402, 202, 302, 30, 'Chennai Warehouse', '01-SEP-2026');

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (403, 203, 303, 40, 'Coimbatore Warehouse', '02-SEP-2026');

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (404, 204, 304, 20, 'Bangalore Warehouse', '02-SEP-2026');

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (405, 205, 305, 35, 'Chennai Warehouse', '03-SEP-2026');

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (406, 201, 306, 15, 'Madurai Warehouse', '03-SEP-2026');

1 row created.

```

6. Seller Product Information

```
SQL> SELECT
  2     s.Seller_Name,
  3     p.Product_Name,
  4     i.Quantity,
  5     i.Warehouse,
  6     i.Last_Updated
  7 FROM Seller s
  8 JOIN Inventory i
  9     ON s.Seller_ID = i.Seller_ID
 10 JOIN Product p
 11     ON i.Product_ID = p.Product_ID
 12 ORDER BY s.Seller_Name;
```

SELLER_NAME

PRODUCT_NAME

QUANTITY

WAREHOUSE

LAST_UPDA

Fashion Distributors
Cotton Shirt
15

SELLER_NAME

PRODUCT_NAME

QUANTITY

WAREHOUSE

LAST_UPDA

Madurai Warehouse
03-SEP-26

SELLER_NAME

PRODUCT_NAME

QUANTITY

WAREHOUSE

LAST_UPDA

Fossil Watch Suppliers
Analog Watch
20

SELLER_NAME

PRODUCT_NAME

QUANTITY

WAREHOUSE

7.Inventory Status

7.1 Display Available Products

```
SQL> SELECT
2      p.Product_ID,
3      p.Product_Name,
4      i.Quantity,
5      i.Warehouse
6 FROM Product p
7 JOIN Inventory i
8      ON p.Product_ID = i.Product_ID
9 WHERE i.Quantity > 0;
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
QUANTITY
-----
WAREHOUSE
-----
      201
Cotton Shirt
      50
Chennai Warehouse
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
QUANTITY
-----
WAREHOUSE
-----
      202
Floral Dress
      30
Chennai Warehouse
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
QUANTITY
-----
WAREHOUSE
-----
      203
Running Shoes
      40
Coimbatore Warehouse
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
QUANTITY
-----
WAREHOUSE
-----
      204
Analog Watch
      20
```

7.2 Display Unavailable Products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Quantity,
  5     i.Warehouse
  6 FROM Product p
  7 JOIN Inventory i
  8     ON p.Product_ID = i.Product_ID
  9 WHERE i.Quantity = 0;

no rows selected

SQL>
```

8. Inventory Update

8.1 Update Inventory Quantity

```
SQL> UPDATE Inventory
  2 SET Quantity = 30,
  3     Last_Updated = '05-SEP-2026'
  4 WHERE Inventory_ID = 404;

1 row updated.

SQL> |
```

8.2 Check the updated record

```
SQL> SELECT * FROM Inventory
  2 WHERE Inventory_ID = 404;

INVENTORY_ID PRODUCT_ID  SELLER_ID  QUANTITY
-----
WAREHOUSE
-----
LAST_UPDA
-----
          404          204          304          30
Bangalore Warehouse
05-SEP-26
```

9. Inventory Status Reports

9.1 Complete Inventory Report

```

SQL> SELECT
  2     i.Inventory_ID,
  3     s.Seller_Name,
  4     p.Product_Name,
  5     i.Quantity,
  6     i.Warehouse,
  7     i.Last_Updated
  8 FROM Inventory i
  9 JOIN Seller s
 10     ON i.Seller_ID = s.Seller_ID
 11 JOIN Product p
 12     ON i.Product_ID = p.Product_ID
 13 ORDER BY i.Inventory_ID;

```

INVENTORY_ID

SELLER_NAME

PRODUCT_NAME

QUANTITY

WAREHOUSE

LAST_UPDA

401

INVENTORY_ID

SELLER_NAME

PRODUCT_NAME

QUANTITY

WAREHOUSE

LAST_UPDA

Levis Suppliers

INVENTORY_ID

SELLER_NAME

PRODUCT_NAME

QUANTITY

WAREHOUSE

LAST_UPDA

Cotton Shirt

INVENTORY_ID

SELLER_NAME

PRODUCT_NAME

QUANTITY

9.2 Available and Unavailable Product Count

```
SQL> SELECT
  2     CASE
  3         WHEN Quantity > 0 THEN 'Available'
  4         ELSE 'Unavailable'
  5     END AS Stock_Status,
  6     COUNT(*) AS Total_Products
  7 FROM Inventory
  8 GROUP BY
  9     CASE
 10         WHEN Quantity > 0 THEN 'Available'
 11         ELSE 'Unavailable'
 12     END;

STOCK_STATU TOTAL_PRODUCTS
-----
Available          6

SQL>
```

9.3 Total Stock by Seller

```
SQL> SELECT
  2     s.Seller_Name,
  3     SUM(i.Quantity) AS Total_Stock
  4 FROM Seller s
  5 JOIN Inventory i
  6     ON s.Seller_ID = i.Seller_ID
  7 GROUP BY s.Seller_Name
  8 ORDER BY s.Seller_Name;

SELLER_NAME
-----
TOTAL_STOCK
-----
Fashion Distributors
      15

Fossil Watch Suppliers
      30

Levis Suppliers
      50

SELLER_NAME
-----
TOTAL_STOCK
-----
Nike Sports Suppliers
      40

Vero Moda Suppliers
      30

Wildcraft Suppliers
      35

6 rows selected.
```

9.4 Products with Zero Stock

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Quantity,
  6     i.Warehouse
  7 FROM Product p
  8 JOIN Inventory i
  9     ON p.Product_ID = i.Product_ID
 10 JOIN Seller s
 11     ON i.Seller_ID = s.Seller_ID
 12 WHERE i.Quantity = 0;

no rows selected
```

